Araveeti Sai Tejeswar Reddy

ce24btech11007@iith.ac.in | tejeswar2220@gmail.com | +91 8179771767 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2024	8.33
Narayana inter college	Class XII (Telangana State Board of Intermediate Education)	2024	97.20%
VIswabharathi WISEWOODS EM School	Class X (Central Board of Secondary Education)	2022	93.40%

EXPERIENCE

Progressive Collapse of Steel Frames under Impulsive Loading | Abaqus, Finite Element Analysis (FEA)

- Developed **finite element models in Abaqus** to investigate steel-frame response under impulsive loading, incorporating material behaviour, geometric effects, and varying loading conditions.
- Evaluated **deformation, stress distribution, and failure mechanisms** through nonlinear dynamic simulations to identify critical stages of structural deterioration.
- Evaluated **deformation, stress distribution, and failure mechanisms** through nonlinear dynamic simulations to identify critical stages of structural deterioration.

SKILLS

Technical Tools: Abaqus, AutoCAD, SQL, Microsoft Excel, Microsoft PowerPoint

Engineering & Analysis: Finite Element Analysis (FEA), Structural Analysis, Structural Simulation, Dynamic Analysis, Failure Analysis

Engineering Fundamentals: Mechanics of Solids, Engineering Mechanics, Fluid Mechanics, Hydraulic Engineering, Steel Structures

Data & Problem Solving: Data Analysis, Exploratory Data Analysis, Statistical Analysis, Design Optimization, Technical Problem Solving

Professional Skills: Leadership, Technical Communication, Cross-functional Collaboration, Project Planning, Stakeholder Management

PROJECTS

PulsePay | MySQL,Statistical Testing, Power BI

- Analyzed **5,150 KYC attempts** using SQL to identify a **21.04 percentage-point drop in KYC completion** on Android 5.4 and investigated document-upload errors to identify the root cause.
- Evaluated **6,032-user KYC A/B experiment** using statistical testing, analyzing conversion lift, confidence intervals, and experiment integrity to support product decisions.
- Analyzed **activation, D30 retention, acquisition quality, and user behavior** using SQL and Power BI to identify product improvement opportunities and translate insights into actionable recommendations

Design & Fabrication of a Working Pelton Wheel| Hydraulic Engineering

- Designed and fabricated a **working Pelton wheel model** by applying fluid mechanics and hydraulic engineering principles, focusing on impulse turbine operation and efficient conversion of hydraulic energy into mechanical energy.
- Applied **engineering design, material selection, fabrication, and assembly** techniques to develop a functional turbine model, while studying water-jet interaction with buckets and its influence on turbine performance

- Tested and evaluated the fabricated model under operating conditions to assess **water-jet impact, mechanical output, stability, and overall performance**, refining design parameters to improve the functionality and reliability of the system.

POSITION OF RESPONSIBILITY

Elan & nVision | Hospitality & Transport Head

- Led a **30+ member cross-functional team** managing end-to-end **accommodation, transportation, and hospitality operations** for **35,000+ participants, guests, and artists**, coordinating with multiple teams to ensure seamless event execution.
- Conceptualized and led a **flagship workshop generating 4 lakh in revenue**, overseeing **planning, budgeting, marketing coordination, participant engagement, and on-ground execution**.

Civil Engineering Society (CES), IIT Hyderabad | Executive

- Planned and executed **hospitality, event logistics, seminars, workshops, and technical events** by coordinating with faculty, vendors, speakers, and student volunteers.
- Managed **venue preparation, participant support, scheduling, and resource allocation**, collaborating with cross-functional teams to ensure smooth event execution and an organized participant experience.

Class Representative | IIT Hyderabad (2024-2025)

- Led **communication and coordination for 100+ students**, addressing academic and administrative concerns while facilitating effective collaboration between students and faculty.
- Coordinated **class-wide academic activities**, including schedule updates, examinations, course registrations, and institute announcements, while facilitating stakeholder communication and ensuring smooth execution of academic processes.

RELEVANT COURSES

Mathematics: Calculus I & II, Linear Algebra, Differential Equations, Probability, Statistics, Transform Techniques, Complex Variables

Core Civil: Geotechnical Engineering, Mechanics of Solids, Structural Analysis, Engineering Drawing, Construction Materials & Management, Environmental Systems Engineering,Concrete & Steel Design ,Hydraulics, Highway Design.

EXTRACURRICULAR

Inter IIT Sports Meet (IISM) | Coordinator -- Coordinated with students across IITs during the Inter IIT Sports Meet.

Vidhyadhan, NSS | Volunteer -- Conducted interactive learning sessions for students at government schools.